

Climate Impact Analysis of Bat Lake Salamanders (*A. maculatum*)

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This report describes an analytical pipeline forecasting a northern population of yellow spotted salamanders over the remainder of the 21st century. The analysis exists as part of a larger research paper (Kaufman et al., 20XX), and much of the text, figures, and tables in this report were taken directly from there. All code and data used to complete this sub-analysis are available at <https://github.com/lepasserin/a-maculatum-climate-impact>.

1 Abstract (from pre-print)

Temperate regions experience pronounced interannual climatic variability, providing the opportunity to understand the drivers of temporal variation in survival, and to model future impacts of environmental change. Temperate amphibians are sensitive to climatic variation and are experiencing dramatic declines in range size and abundance under global environmental change. We developed a priori hypotheses for climate effects on amphibian survival, then combine fifteen years of mark-recapture data from a population of spotted salamanders (*Ambystoma maculatum*) with Bayesian hierarchical modelling to test the association between environmental variables and annual survival. We found that greater temperatures during the spring inbound breeding migration led to a 50% decline in annual survival, while greater precipitation during the outbound migration was associated with a survival increase of 30%. Evaporative water loss may be the major driver of survival variation during the spring migration.

We then projected population abundance under three climate change scenarios, incorporating the environmental effects on annual survivorship that were discovered. In the mildest scenario, projected increases in spring precipitation were able to compensate for their warmer temperatures, resulting in an approximately stable population size. Faced with more extreme future climates, the population declines to over half of its initial size. Our study draws urgent attention to the migratory dynamics of temperate, spring breeding amphibians faced with a changing climate.

2 Methods (from pre-print)

We projected the abundance of the Bat Lake spotted salamander population through the remainder of the 21st century, under three different climate scenarios. First, we forecasted the inbound temperature and outbound precipitation covariates from the breeding migration model using climate data from global circulation models (GCMs). Using the fitted model, these forecasts were used to estimate future adult female survivorship values. Finally, these were embedded into a population growth model to project the population's abundance through time. Our pipeline is summarized in Figure 1.

at year's mean April temperature (leveraging its strong, negative
out date; Supplementary Figure 2), and was set to end 22 days
the average breeding season length observed during the BLISS
; validated by comparing hindcasted and observed environment

Figure 1: Schematic representation of the climate impact analysis. Vector quantities are bolded. Theta represents constant life-history parameters, derived from literature.

Estimates of mean temperature, minimum temperature and total precipitation were required for each day between March–May for the years 2005–2100 to correctly calculate the migration

model’s environmental covariates. These data were sourced from the Coupled Model Inter-comparison Project Phase 6 (CMIP6) dataset, which provides daily weather predictions that have been statistically downscaled to spatial resolutions of 10 km² (Eyring et al. 2016). We used the closest available grid point to the Algonquin East Gate Climate Station (45° 32’ N, 78° 17’ W), approximately 2.2 km away. For each climate variable the ensemble mean of the 26 available GCMs was used. Three such datasets were compiled, one for each of three Shared Socio-economic Pathways (SSPs) representing conservative (SSP1), moderate (SSP2) and extreme (SSP5) fossil-fueled development respectively (Riahi et al. 2017).

We applied two statistical bias corrections to the GCM data, in accordance with standard practice for climate impact analysis (Sharma et al. 2007; Hagemann et al. 2011). For mean and minimum temperature, additive linear scaling was used to correct for long-run bias in mean and minimum temperature data, following the procedure described in Hempel et al. (2013). For total precipitation, local intensity scaling was used to correct the intensity and frequency of rainy days, applied as described in Smitha et al. (2018). These procedures were applied to the GCM data over a 20-year period (2005-2024) where their results could be directly compared to the Algonquin East Gate Climate Station’s historical records. Once suitable correction factors had been identified, these corrections were applied to future GCM data (see Appendix A for explicit correction equations and a visual comparison of raw versus bias-corrected simulated data).

Once bias-corrected, GCM data was used to predict the salamander breeding season start and end dates, enabling the calculation of the specific environmental covariates used in the CJS model, for each year through to 2100. For a given year, the breeding season start date was estimated using that year’s mean April temperature (leveraging its strong, negative correlation with Bat Lake ice out date; Appendix B), and was set to end 22 days later, corresponding to the average breeding season length observed during the BLISS study period. These results were validated by comparing hindcasted and observed environmental indices over our 2008-2022 study period (Appendix B). Using this time interval, annual survivorship was estimated using the breeding migration model covariates for inbound temperature and

outbound precipitation. This process was repeated for each of the three SSPs, and results were validated by comparing hindcasted and observed environmental indices over the 2008-2022 BLISS study period (Appendix B).

Finally, we used values of predicted adult survivorship from our breeding migration model within a population growth model to estimate long-term population trajectories under the different climate scenarios. Our population model is an age-structured “Leslie” matrix model, modified so that the final class represented all reproductively mature adults (Smith and Trout 1994). We chose this model over a simpler one to account for the time lag between changes in adult survivorship and their effects on future adult recruitment. Since female spotted salamanders mature at around 6 years old (Boyle, 2012), this resulted in a seven-staged discrete growth model. We sourced mature fecundity estimates for the Bat Lake population from Moldowan (2023). As adequate estimates of annual juvenile survivorship could not be determined, they were set such that the resulting model had a finite rate of increase of 1 (Congdon et al. 1994). The output of the fitted CJS model evaluated at the average values of all significant environmental variables (i.e. the model intercept) was used as the baseline mature survivorship (see Appendix C for full model parameterization). Simulations were initialized in the year 2019 at the base model’s stable age distribution with 1840 starting adult females. Simulations initialized at different initial abundances and starting years had qualitatively similar long-term trajectories (Appendix C).

Appendix

A Statistical Bias Corrections

The daily projections of mean temperature, minimum temperature, and total precipitation used in this study (hereafter referred to as “climate projections”) were sourced originally from the Coupled Model Intercomparison Project Phase 6 (CMIP6) dataset, which provides daily estimates of these climate variables under various socio-political scenarios (called “Shared Socioeconomic Pathways”, often abbreviated as SSPs). Since GCM output is coarse-grained, *Environment and Climate Change Canada* provides downscaled estimates with higher spatial resolution [10 km²; Eyring et al. (2016)], for each of three SSPs (SSP1, SSP2, and SSP5 ; see (Riahi et al. 2017) for detailed definitions). These are more appropriate for the analysis of highly localized populations such as the Bat Lake spotted salamander population. Even with downscaling, however, climate projections are often still systematically biased (Sharma et al. 2007; Hagemann et al. 2011); while overall trends are often quite accurate, projected estimates may deviate substantially from observations in their absolute values and/or interannual variations.

We implemented two bias correction methods to correct specific, systematic deviations between observed and simulated climate metrics. In both cases, these methods were first applied over a historical period (2005 to 2024) where observed daily climate data was available near our study site (measurements taken from Algonquin East Gate Climate Station), to determine suitable correction factors. Then, these correction factors were applied to correct bias in future projections (through to 2100), under the assumption that systematic bias in the past will manifest itself in the same way in the future. We note here that projections from all three SSPs considered were corrected using the same coefficients.

A.1 Correcting Temperature Estimates

For both mean and minimum daily temperature, downscaled estimates showed similar trends to observed measurements over the entire historical period, but with consistently higher monthly averages (see Figure A1 and Figure A2). Simulated temperatures also exhibited smaller day-to-day variation compared to observed trends, likely because our simulated estimates are based on the ensemble mean of 26 different GCMs, which artificially deflates variance. Methods that correct the variance and the mean in simulated temperature data simultaneously exist, but have been shown to impact the climate change signal in certain datasets (Hagemann et al. 2011). Given how temperature projections are used in further analyses, we opted to only correct the systematic bias in the average, preserving the climate change signal most faithfully. To this end we applied additive linear scaling following Hempel et al. (2013), implemented as described below.

Let i represent the March of a given year between 2005 and 2024. Then, let T_{ij}^{OBS} represent the observed temperature (mean or minimum) of the j^{th} day in that month respectively. T_{ij}^{SIM} and T_{ij}^{ADJ} are defined similarly, for simulated (SIM) and bias-corrected (ADJ) daily values. We calculate the linear scaling constant C as follows:

$$C = \left(\sum_{i=2005}^{2024} T_i^{OBS} - T_i^{SIM} \right) / 20 \quad (1)$$

where T_i^{OBS} and T_i^{SIM} represent the average temperature of the i^{th} March from 2005 to 2024, for observed and simulated data respectively. C can then be used to scale future daily temperature simulations like so:

$$T_{i,j}^{ADJ} = C + T_{i,j}^{SIM} \quad (2)$$

Notice that C is calculated over the entire period, and not per-year. This allows us to apply the correction shown in Equation 2 to every March in the forecasted simulated data. This

procedure can then be repeated for April and May, mean and minimum temperatures separately, resulting in the estimation of 6 linear scaling constants to correct GCM-derived temperature forecasts.

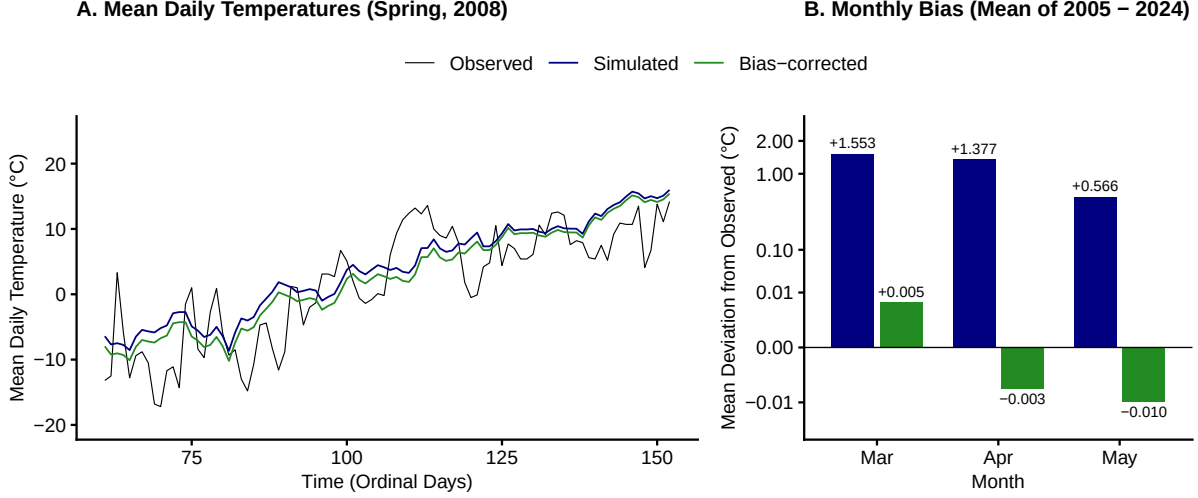


Figure A1: (A) Time series of observed (black), simulated (blue), and bias-adjusted (green) mean daily temperatures during the spring of 2008 (year chosen arbitrarily, for illustrative purposes). (B) Systematic deviation of simulated (blue) and bias-adjusted (green) estimates from observed monthly mean temperature, averaged from 2005 to 2024. Year-months with more than 20% missing observations are excluded, matching the rule used to fit the correction factors.

A.2 Local Intensity Scaling

For daily total precipitation, simulated data had similar monthly and seasonal averages as observed data, but grossly underestimated the day-to-day variance in precipitation, effectively predicting a drizzle each day. This resulted in an underestimated number of rainy days, and an underestimated intensity of wet days. Since the precipitation main effect used in our analysis is sensitive to the number of rainy days (see main text), this drizzle bias required correction. The method used, local intensity scaling, is described below.

Let i represent the March of a given year between 2005 and 2024. Then, let P_{ij}^{OBS} represent the observed precipitation (total) on the j^{th} day in that month respectively. P_{ij}^{SIM} and P_{ij}^{ADJ}

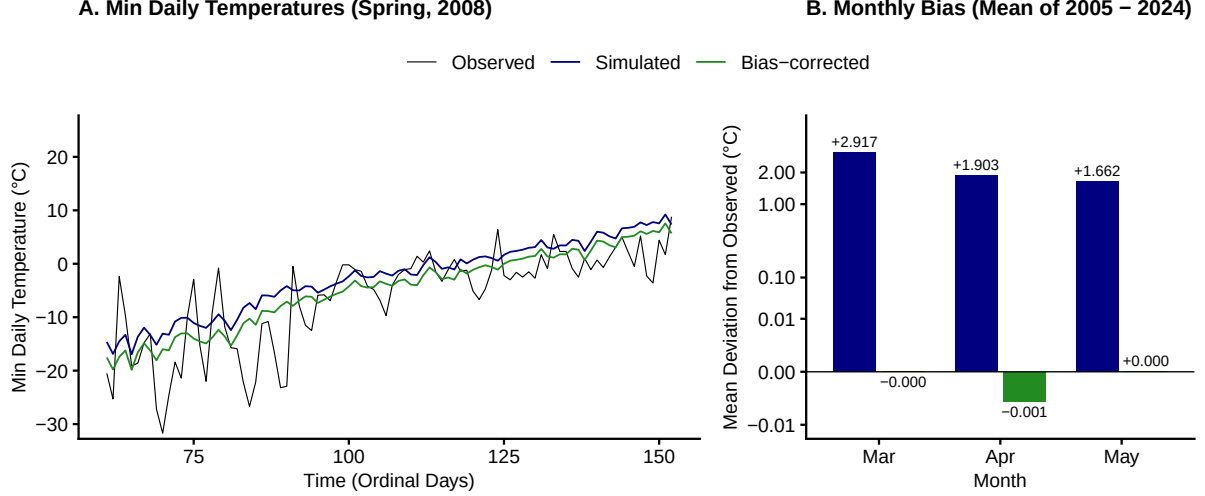


Figure A2: (A) Time series of observed (black), simulated (blue), and bias-adjusted (green) minimum daily temperatures during the spring of 2008 (year chosen arbitrarily, for illustrative purposes). (B) Systematic deviation of simulated (blue) and bias-adjusted (green) estimates from observed monthly minimum temperature, averaged from 2005 to 2024. Year-months with more than 20% missing observations are excluded, matching the rule used to fit the correction factors.

are defined similarly, for simulated (SIM) and bias-corrected (ADJ) daily values. First, a fixed threshold t is determined such that the average number of wet days with simulated precipitation totals below t is equal to the average number of observed dry days. This constant is used to correct for the number of dry days in the simulated data. Then, we calculate the observed average precipitation among wet days (D_1), and the average precipitation among simulated days above t (D_2). Their ratio, $D = \frac{D_1}{D_2}$, is used to correct for the expected intensity of wet days in the simulated data. These constants are then applied to the simulated precipitation data as follows:

$$P_{i,j}^{ADJ} = \begin{cases} 0, & \text{if } P_{i,j}^{SIM} < t \\ (P_{i,j}^{SIM} - t) \times D, & \text{otherwise} \end{cases}$$

Similarly to temperature, three sets of (D, t) pairs were fit, for each month in our dataset.

A.3 Bias Correction Results & Diagnostics

All fitted correction factors are shown in Table A1. Monthly biases in simulated temperatures were more extreme earlier in the season, and for minimum temperatures. Simulated precipitation days were deemed dry when their total precipitation did not exceed around 2mm, representing a light shower. Simulated wet days saw their total precipitation amount inflated by a factor of 4 to 7, more intense later in the season. Figures 1.1, 2.1, and 3.1 illustrate the effect of the correction procedures compared to observed data, for each climate variable.

Table A1: Fitted correction factors applied to simulated climate projections

Correction Factor	Value
March mean temperature offset (°C)	-1.54834
April mean temperature offset (°C)	-1.38055
May mean temperature offset (°C)	-0.57552
March minimum temperature offset (°C)	-2.91662
April minimum temperature offset (°C)	-1.90356
May minimum temperature offset (°C)	-1.66153
March dry day threshold (mm)	2.11449
April dry day threshold (mm)	2.16254
May dry day threshold (mm)	2.57951
March intensity coefficient (unitless)	4.03366
April intensity coefficient (unitless)	5.97328
May intensity coefficient (unitless)	6.87419

B Visually Validating Environmental Index Forecasts

Bias-corrected climate variables were used to forecast the following annual variables: - Breeding season start date (from which the breeding season end date was deterministically derived) -

Inbound temperature index (see definition in main text) - Outbound precipitation index (see definition in main text)

Here we present our forecasted indices, for the period from 2008 (start of the BLISS project) to 2100 (end of available GCM estimates). Three forecasts were made in each case, one for each of the three climate scenarios considered. Shown concurrently are the observed indices, available during the BLISS project study period (2008 to 2022). These are shown in Figures 2.1, 2.2, and 2.3. In each case, the simulated projection is similar to the mean of the observed index during the BLISS study period. However, the true year-to-year variance in these indices was always larger than what was produced from the GCM data, apart from forecasted precipitation indices, which matched observed variance quite closely.

C Population Growth Model Specification

We used a modified age-structured population growth model to forecast the female Bat Lake spotted salamander population through time. Let N_t be a vector of length seven representing the population of salamanders, binned into age groups, in year t . Then, the population in the following year (N_{t+1}) is calculated as follows:

[Equation]

Population growth in our model is governed primarily by the annual adult female survivorship term $S_a(t)$, which varies from year to year according to those environmental characteristics which correlated with survivorship in the CJS survivorship model fit in the main paper. We write:

[Equation]

where $\text{Temp}_{t,j}$ and $\text{Precip}_{t,j}$ are the standardized temperature and precipitation indices (see main text for full definitions) of Bat Lake in the year t and climate scenario j . β_i are the fitted coefficients from the CJS survivorship model. $S_a(t)$ estimates for every scenario for each year from 2000 to 2100 are shown visually in Figure 3.1.

Population growth also depends on three other life-history parameters which were held constant over time. S_m denotes the subadult (i.e. metamorph) annual survivorship, which is shared between all subadult age classes, p denotes the probability that a mature female skips reproduction in a year, and f denotes the average number of offspring which survive until the next year produced by a reproducing female. Reproductive skipping is a well-documented component of yellow spotted salamander life-history. All parameter estimates are reported in Table 3.1. While skipping and fecundity estimates could be found in the literature, subadult survivorship could not. As such, the parameter was numerically optimized such that the resulting model had a finite rate of increase λ near 1, when $S_a(t) = \text{logit}^{-1}(\beta_0) \approx 0.72$ (i.e., adult female survivorship assuming study-average temperature and precipitation indices). We thought this decision sensible since the population did not show signs of increase or decrease during the BLISS study period (2008 to 2024). This baseline survivorship was also used to derive the stable-age distribution of the population, using the `eigen.analysis` function from the `popbio` package in R. Projections were initialized in the year 2019 at the base model’s stable age distribution with 1840 starting adult females.

We tested the robustness of our population projections by varying the following elements of our model and comparing the output with the base run:

- Different combinations of the non-focal parameters p , f (optimizing S_m such that λ is near 1 in each case).
- Earlier starting year (checked from 2000 to 2025).

The results of these robustness tests are displayed in Figures 3.2 and Figure 3.3 respectively, and show that our projections are not sensitive to changes in either case. On top of these computational verifications, we know mathematically that varying the absolute starting abundance does not change the relative model dynamics at all.

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